Module 1 Homework 2 Challenge

Provide the best answer for each of the following questions.

Time Remaining 00:59:16

Main Question Set

Solve six of the first ten questions to achieve a passing score of 60%. Don't miss the learning opportunity in the rest of the problems, though -- spend the full time and attempt every one of them!

1

Review Later

An alternative recipe for cookies requires

$$\begin{array}{l} 8 \text{ eggs} \\ 6 \text{ cups flour} \\ 3 \text{ cups sugar} \end{array}$$

If you want to use exactly 5 eggs to make these cookies, what will be the adjusted recipe for these cookies?

- $\begin{array}{l} 5 \text{ eggs} \\ 5 \text{ cups flour} \\ 5 \text{ cups sugar} \end{array}$
- $\begin{array}{l} 5 \text{ eggs} \\ 3\frac{3}{4} \text{ cups flour} \\ 1\frac{7}{8} \text{ cups sugar} \end{array}$
- $\begin{array}{l} 5 \text{ eggs} \\ 2 \text{ cups flour} \\ 5 \text{ cups sugar} \end{array}$
- $\begin{array}{l} 5 \text{ eggs} \\ 6 \text{ cups flour} \\ 6 \text{ cups sugar} \end{array}$
- $\begin{array}{l} 5 \text{ eggs} \\ 5 \text{ cups flour} \\ 2 \text{ cups sugar} \end{array}$

2

Review Later

The ratio of red candies, to blue candies, to yellow candies in a bag is $2:3:5$. What percent of the total candies are red?

- 20%
- $\frac{1}{3}$
- 10%
- 66.67%
- 2%

3

Review Later

Which of the following sequences is a geometric sequence? In a geometric sequence, a term is found by multiplying the previous term by a certain number, called a factor, which is the same for all terms.

- $10,\ 110,\ 1110,\ 11110$
- $1,\ 2,\ 3,\ 4$
- $108,\ 36,\ 12,\ 4$
- $0.1,\ 0.11,\ 0.111,\ 0.1111$
- $12,\ 14,\ 16,\ 18$

4

Review Later

An oak tree is 80 feet taller than a flagpole, and three times as tall as it. How many feet tall is the flagpole? Your answer should be the number only, without units.

5

Review Later

Anna has a water bottle that is half full. After she drinks 15 fluid ounces from the bottle, it is only one-fifth full. What is the total capacity of the water bottle, in fluid ounces?

6

Review Later

Joe wants to buy a shirt that costs $\$50$, and he has a coupon for 30% off. When he arrives at the store, however, he sees that the shirt has already been discounted 20%, but he can still use the coupon. How much does Joe end up paying for the shirt, in dollars?

7

Review Later

If two similar rectangles are proportional to each other, with the first rectangle having dimensions length_1 and width_1, and the second rectangle having dimensions length_2 and width_2, then all of the following statements are true EXCEPT for which one?

- $\frac{\text{length}_2}{\text{length}_1} = \frac{\text{width}_2}{\text{width}_1}$
- $\frac{\text{length}_2}{\text{width}_1} = \frac{\text{width}_2}{\text{length}_1}$
- $\frac{\text{length}_2}{\text{width}_2} = \frac{\text{length}_1}{\text{width}_1}$
- $\frac{\text{width}_2}{\text{length}_2} = \frac{\text{width}_1}{\text{length}_1}$
- $\frac{\text{width}_2}{\text{width}_1} = \frac{\text{length}_2}{\text{length}_1}$

8

Review Later

Adam eats 20% of a pizza, Bethany eats another $\frac{1}{3}$ of the pizza, and Charlie eats the remaining 14 slices. How many slices total did the pizza originally have?

9

Review Later

Which of the following is the same as 2% of 50?

- 20% of 5

- 5% of 20
- $\frac{2}{100} \times 50$
- 50% of 2
- All of the above

10

Review Later

Your brother is half as tall as a tree which is three times as tall as you. What percent taller is your brother compared to you?

- 30%
- 300%
- 50%
- $\frac{1}{2}$
- 3.5%

Bonus Challenge Questions

These optional questions will not count towards your final score. However, we strongly suggest trying them for the benefit of the challenge!

11

Review Later

By what percent will the area of a rectangle increase if its sides increase by 20%? Your answer should be a number, without the $\%$ sign.

12

Review Later

You paid $\$15.73$ total for a meal at a restaurant, which includes the cost of the tip, which is 15% of the menu price of the meal. What is the menu price of your meal?

- $\$15.73 \times 0.15$
- $\$15.73 \times 1.15$
- $\$15.73 \times 0.85$
- $\frac{\$15.73}{0.15}$
- $\frac{\$15.73}{1.15}$

13

Review Later

A brother's age is 40% of his sister's age. What percent of the brother's age is the sister's age? Your answer should be the number only, without units.

14

Review Later

Calculate $1001 \times 0.0142857142857142857\ldots$ as a *(terminating)* decimal.

15

Review Later

Number B is twice as big as Number C, and Number A is $\frac{5}{6}$ of Number B. Find the percent of Number A that is Number C. Your answer should be the number only, without units.

16

Review Later

At the beginning of the year, the number of men employed at a plant was 40% of the total factory workers. After six months, 6 men were hired and 5 women quit, making the number of men and women at the factory equal. How many people were working at the plant at the beginning of the year?

17

Review Later

Alice and Bob are playing a game together. If Bob uses a fair dice, there is a 50% chance that he will win. However, there is a 60% chance that Bob will cheat using an unfair dice which gives him a 75% chance of winning. What percent of the time does Bob actually win? Your answer should be the number only, without units.

Super Bonus Questions

These questions are an extreme challenge! Some of the most interesting, creative and clever questions are the hardest ones. So try every single one. Even if you can't figure it out, you'll still be deepening your understanding of the important math concepts that underlie this course.

18

Review Later

A watermelon is 99% water. You have 100 pounds of watermelon. After a week under the sun, the watermelon has dried down to being 98% water. What is the total weight of the watermelon now? Your answer should be the number only, without units.

19

Review Later

Evan is deciding whether to open a bank account in Bank A or Bank B. Bank A has a yearly annual interest rate of 10% (that is, at the end of every year, the amount of money in a Bank A account increases by 10%. Bank B has an interest rate that is applied once every three years (that is, at the end of every three-year period, the amount of money in a Bank B account increases by some fixed rate). What is the smallest such interest rate that Bank B must have in order to be competitive with Bank A?

Express your answer as a percent rounded to the nearest tenth, e.g. 50.5

20

Review Later

Near a lake, 17% of the pollywogs eat 75% of the flies. If they had each eaten 7% more, what percent less do the rest of the pollywogs eat?

View Marked Questions

Finished

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